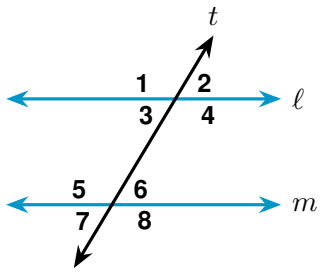
Parallel Lines & Transversals — Practice — Answer Key

A worked example comes before each type. Show your thinking.

Name: _____ Date: _____

Type 1 Find the Angles

Watch Me — Worked Example



$\ell \parallel m$ and $m\angle 2 = 65°$. Find $m\angle 7$.
$\angle 2$ and $\angle 7$ are alternate exterior, so they are congruent. **Answer: $65°$.**
Find $m\angle 8$: $\angle 7$ and $\angle 8$ are a linear pair, so $180 - 65 = 115°$.

In every problem $\ell \parallel m$ (figure above).

1. $m\angle 1 = 120°$. Find $m\angle 5$.

$120°$

2. $m\angle 4 = 58°$. Find $m\angle 5$.

$58°$

3. $m\angle 3 = 105°$. Find $m\angle 5$.

$75°$

4. $m\angle 6 = 72°$. Find $m\angle 3$.

$72°$

5. $m\angle 8 = 131°$. Find $m\angle 1$.

$131°$

6. $m\angle 2 = 47°$. Find all eight angles.

$47°$: 2, 3, 6, 7; $133°$: 1, 4, 5, 8

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Type 2 Solve With Algebra: Congruent Pairs

Watch Me — Worked Example

Corresponding angles: $(4x-10)^\circ$ and $(2x+30)^\circ$. Congruent: $4x-10=2x+30 \Rightarrow 2x=40 \Rightarrow x=20$. Angle: $4(20)-10=70^\circ$. **Answer: $x=20,\ 70^\circ$**

7. Alternate interior angles: $(5x-8)^\circ$ and $(3x+22)^\circ$. Find x and the angle.

$x = 15,\ 67^\circ$

8. Alternate exterior angles: $(7x+4)^\circ$ and $(9x-20)^\circ$. Find x and the angle.

$x = 12,\ 88^\circ$

Type 3 Solve With Algebra: Supplementary Pairs

Watch Me — Worked Example

Same-side interior angles: $(3x+20)^\circ$ and $(2x+10)^\circ$. They add to 180°: $5x+30=180 \Rightarrow x=30$. Angles: $3(30)+20=110^\circ$ and $2(30)+10=70^\circ$. Check: $110+70=180$ ✓

9. Same-side interior angles: $(4x+12)^\circ$ and $(2x-6)^\circ$. Find both angles.

$x = 29$: 128° and 52°

10. $m\angle 1=(6x+4)^\circ$ and $m\angle 7=(2x+16)^\circ$. Find x.

$x = 20$

Type 4 Two Steps

Watch Me — Worked Example

$m\angle 1=(x+40)^\circ$ and $m\angle 6=(2x+20)^\circ$. $\angle 1 \cong \angle 5$ (corresponding), and $\angle 5$, $\angle 6$ are a linear pair, so $(x+40)+(2x+20)=180 \Rightarrow x=40$.

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11. $m\angle 2 = (3x)^\circ$ and $m\angle 5 = (x + 20)^\circ$. Find x.

$x = 40$

12. HARD $m\angle 4 = (x^2)^\circ$ and $m\angle 5 = (5x + 14)^\circ$, $x > 0$. Find $m\angle 4$.

$x = 7, 49^\circ$

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